

Python Functional Programming Exercises (map, filter, reduce)

Beginner

1. Given a list of numbers, use `map()` to square each element.
2. Use `filter()` to return only even numbers from a list.
3. Combine `map()` and `filter()` to get the cubes of odd numbers only.
4. Use `reduce()` to find the sum of a list.
5. Use `reduce()` to find the maximum number in a list.

Intermediate

6. Given a list of words, use `map()` to return their lengths.
7. Filter out words shorter than 4 letters.
8. Use `reduce()` to concatenate all words into a single string separated by spaces.
9. Given a list of dictionaries representing people, use `filter()` to select only those older than 30.
10. Use `map()` to extract their names.

Advanced

11. Combine `map()`, `filter()`, and `reduce()` to find the sum of squares of even numbers only.
12. Given a list of sentences, use `map()` to split them into words and `reduce()` to count total words.
13. Given a list of tuples (name, score), use `filter()` to keep only scores above 50, and `map()` to get just the names.
14. Use `reduce()` to compute the product of all positive numbers in a list.
15. Given a nested list of numbers, use `reduce()` to flatten it into a single list.